Mathematics for Computer Graphics I

Scalar Fields, Vector Spaces, Affine Spaces, Euclidean Spaces, Dot Product, Cross Product, Coordinate Frames

Carlos Martinho 2026

Scalar Fields

We need to represent size

$\forall \alpha, \beta, \gamma \in S$

commutative:	$\alpha + \beta = \beta + \alpha$	$\alpha \,.\, \beta = \beta \,.\, \alpha$
associative:	$\alpha + (\beta + \gamma) = (\alpha + \beta) + \gamma$	$\alpha \,.\, (\beta \,.\, \gamma) = (\alpha \,.\, \beta) \,.\, \gamma$
distributive:	$\alpha \,.\, (\beta + \gamma) = (\alpha \,.\, \beta) + (\alpha \,.\, \gamma)$	
identity:	$\alpha + 0 = 0 + \alpha = \alpha$	$\alpha.1 = 1.\alpha = \alpha$
inverse:	$\alpha + (-\alpha) = 0$	$\alpha \,.\, (\alpha^{-1}) = 1$

Vector Spaces

We need to represent movement

$\forall \alpha, \beta \in S, \forall u, v, w \in V$

$u + v \in V$

$u + v = v + u$

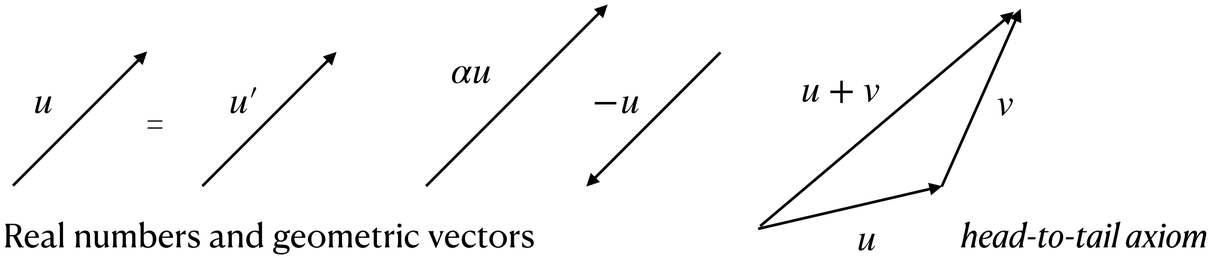
$u + (v + w) = (u + v) + w$

$u + 0 = u$

$u + (-u) = 0$

$\alpha(u + v) = \alpha u + \alpha v$

$(\alpha + \beta)u = \alpha u + \beta u$



Real numbers and geometric vectors

head-to-tail axiom

$u = (u_1, u_2, \ldots, u_n) \in \mathbb{R}^n$
$v = (v_1, v_2, \ldots, v_n) \in \mathbb{R}^n$

$u + v = (u_1 + v_1, u_2 + v_2, \ldots, u_n + v_n)$
$\alpha v = (\alpha v_1, \alpha v_2, \ldots, \alpha v_n)$

n-tuples of scalars (real or complex numbers)

Vector Spaces

Linear combination of vectors

If the only set of vectors v_i such as $\alpha_1 v_1 + \ldots + \alpha_n v_n = 0$ is $\alpha_1 = \ldots = \alpha_n = 0$, then the vectors area **linearly independent**.

The greatest number of independent vectors defines the dimension of the space.

Any set of n linearly independent vectors v_i or v'_i form a **basis** for a space of dimension n, and vectors can be expressed as $v = \beta_1 v_1 + \ldots + \beta_n v_n$ or $v = \beta'_1 v'_1 + \ldots + \beta'_n v'_n$.

There is a $n \times n$ matrix $M_{n \times n}$ such as $\begin{bmatrix} \beta'_1 \\ \vdots \\ \beta'_n \end{bmatrix} = M_{n \times n} \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_n \end{bmatrix}$

Affine Spaces

We need to represent position (points)

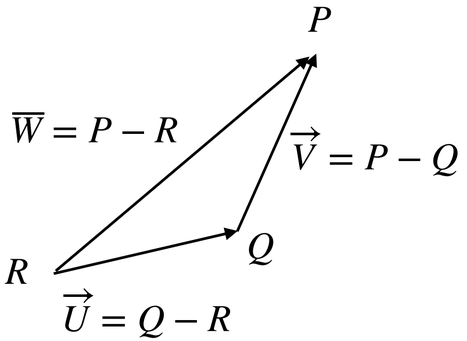
Vectors as point-point subtraction: $\overrightarrow{V} = P - Q, P = Q + \overrightarrow{V}$

Parametric form: $P = Q + \alpha V$

Affine spaces allow the creation of a **frame**: an origin P_0 and a set of n vectors $v_1, \ldots, v_n$ defining a basis.

vector $v = \alpha_1 v_1 + \ldots + \alpha_n v_n$, represented as $(\alpha_1, \ldots, \alpha_n)$

point $P = P_0 + \beta_1 v_1 + \ldots + \beta_n v_n$, represented as $(\beta_1, \ldots, \beta_n)$



$(P - Q) + (Q - R) = (P - R)$

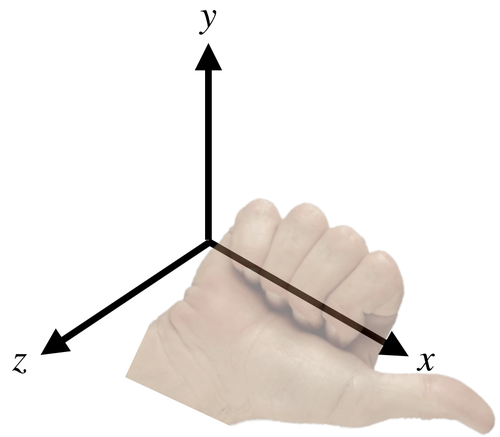
Euclidean Spaces

We need to represent distance and angle

We will work in $\mathbb{R}^3$ and use **right-handed coordinates**.

distance = magnitude of a vector between two points

angle = angle between two vectors "starting" on a same point.



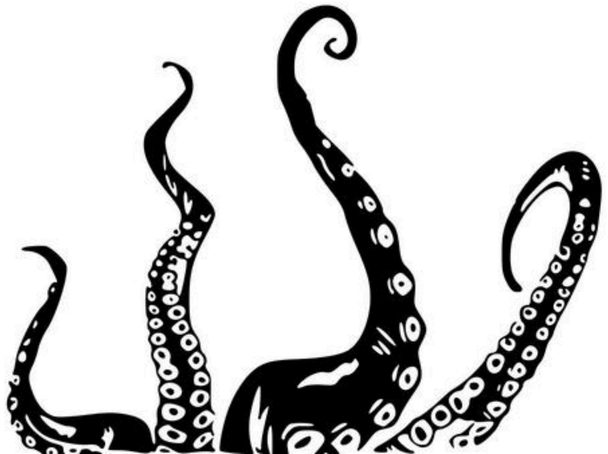
The **cartesian** system is an euclidean space with basis:

$\vec{X} = (1,0,0)$, $\vec{Y} = (0,1,0)$, $\vec{Z} = (0,0,1)$

Dot Product

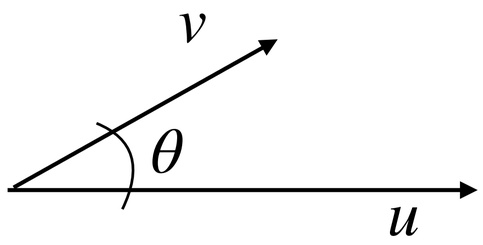
a.k.a. inner product, scalar product





Dot Product

What about vector-vector multiplication?



$u \cdot v = ?$

$u \cdot v = |u||v|\cos\theta$

if $|u| = |v| = 1$, then $u \cdot v = \cos\theta$

$u \cdot v = v \cdot u$

$u \cdot (v \cdot w) = (u \cdot v) \cdot w$

$u \cdot (v + w) = u \cdot v + u \cdot w$

$(\alpha u) \cdot v = \alpha(u \cdot v) = u \cdot (\alpha v)$

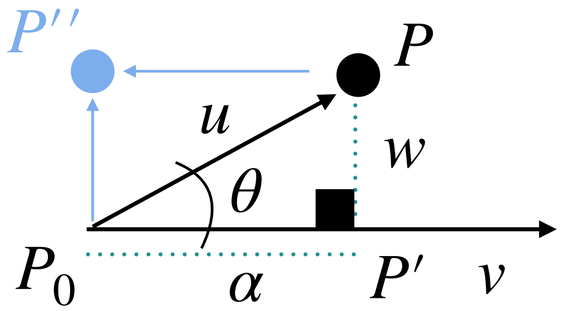
$v \cdot v > 0$ if $v \neq 0, 0 \cdot v = 0$

$|v| = \sqrt{v \cdot v}$

if $u \cdot v = 0$ and $u, v \neq 0$, then $u \perp v$

Dot Product

Examples of use in CG



Angle between u and v :

$$u \cdot v = |u||v|\cos\theta$$

$$\theta = \cos^{-1}\left(\frac{u \cdot v}{|u||v|}\right)$$

Shortest distance to a line from a point :

$$u = \alpha v + w \text{ and } v \cdot w = 0$$

$$u \cdot v = (\alpha v + w) \cdot v$$

$$u \cdot v = \alpha v \cdot v + w \cdot v = \alpha v \cdot v$$

$$\alpha = \frac{u \cdot v}{v \cdot v}$$

where αv is the projection of u on v

Dot Product

Efficient to compute in Cartesian coordinates.

basis: $\vec{X} = (1,0),\ \vec{Y} = (0,1)$

vectors: $\vec{A} = (\alpha_x, \alpha_y) = \alpha_x\vec{X} + \alpha_y\vec{Y},\ \vec{B} = (\beta_x, \beta_y) = \beta_x\vec{X} + \beta_y\vec{Y}$

$$A \cdot B = (\alpha_x\vec{X} + \alpha_y\vec{Y}) \cdot (\beta_x\vec{X} + \beta_y\vec{Y})$$

$$A \cdot B = \alpha_x\beta_x(\vec{X} \cdot \vec{X}) + \alpha_y\beta_y(\vec{Y} \cdot \vec{Y}) + (\alpha_x\beta_y + \alpha_y\beta_x)(\vec{X} \cdot \vec{Y})$$

$$A \cdot B = \alpha_x\beta_x + \alpha_y\beta_y$$

$$A = \begin{bmatrix}\alpha_x \\ \alpha_y\end{bmatrix} \qquad B = \begin{bmatrix}\beta_x \\ \beta_y\end{bmatrix} \qquad A \cdot B = \begin{bmatrix}\alpha_x & \alpha_y\end{bmatrix} \begin{bmatrix}\beta_x \\ \beta_y\end{bmatrix} = A^T B$$

Quadrance

$$Q(v) = v_x^2 + v_y^2 + v_z^2 \textbf{, where } v = (v_x, v_y, v_z)$$

$Q(v) + Q(w) = Q(w - v)$, where $v = (v_x, v_y, v_z)$ and $w = (w_x, w_y, w_z)$

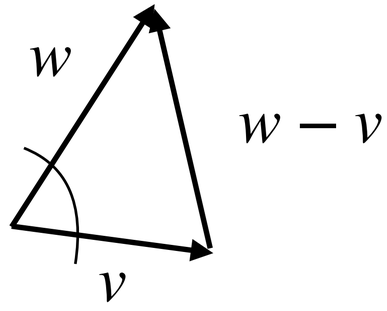
$$v_x^2 + v_y^2 + v_z^2 + w_x^2 + w_y^2 + w_z^2 = (w_x - v_x)^2 + (w_y - v_y)^2 + (w_z - v_z)^2$$

$$v_x^2 + v_y^2 + v_z^2 + w_x^2 + w_y^2 + w_z^2 = (w_x^2 - 2w_xv_x + v_x^2) + (w_y^2 - 2w_yv_y + v_y^2) + (w_z^2 - 2w_zv_z + v_z^2)$$

$$0 = -2w_xv_x - 2w_yv_y - 2w_zv_z$$

$$0 = w_xv_x + w_yv_y + w_zv_z$$

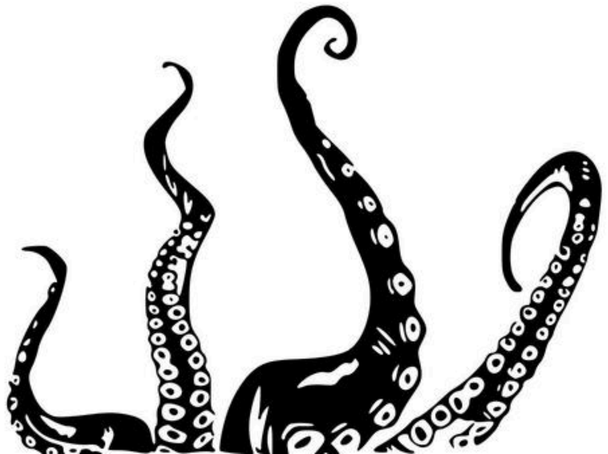
$$0 = w \cdot v$$



Cross Product

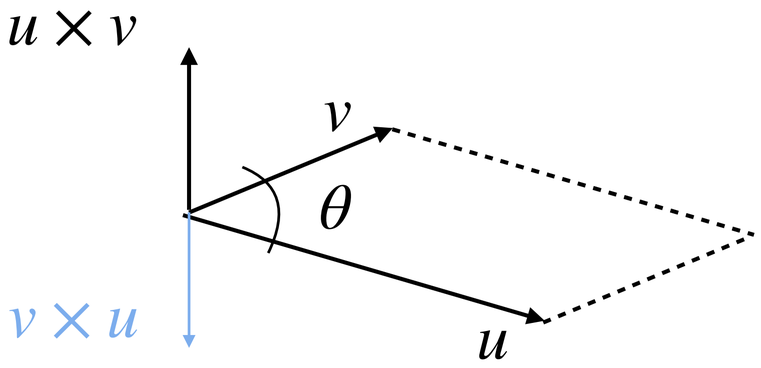
a.k.a. outer product, vector product





Cross Product

What about vector-vector multiplication?



$$|u \times v| = |u||v| \sin\theta$$

$u \times v \perp u, v$ (right-hand rule)

In cartesian 3d space:

$$x \times y = z \qquad y \times z = x \qquad z \times x = y$$

$$y \times x = -z \qquad z \times y = -x \qquad x \times z = -y$$

$u \times w = -w \times u$ (not commutative)

$u \times u = 0$

$u \times (v + w) = u \times v + u \times w$

$u \times (\alpha v) = \alpha(u \times v) = (\alpha u) \times v$

Cross Product

in Cartesian coordinates.

basis: $\vec{X} = (1,0,0),\ \vec{Y} = (0,1,0),\ \vec{Z} = (0,0,1)$

vectors: $\vec{A} = (\alpha_x, \alpha_y, \alpha_z) = \alpha_x\vec{X} + \alpha_y\vec{Y} + \alpha_z\vec{Z},\ \vec{B} = (\beta_x, \beta_y, \beta_z) = \beta_x\vec{X} + \beta_y\vec{Y} + \beta_z\vec{Z}$

$$A \times B = (\alpha_x\vec{X} + \alpha_y\vec{Y} + \alpha_z\vec{Z}) \times (\beta_x\vec{X} + \beta_y\vec{Y} + \beta_z\vec{Z})$$

$$A \times B = \alpha_x\beta_x(\vec{X} \times \vec{X}) + \alpha_x\beta_y(\vec{X} \times \vec{Y}) + \alpha_x\beta_z(\vec{X} \times \vec{Z})$$

$$+\alpha_y\beta_x(\vec{Y} \times \vec{X}) + \alpha_y\beta_y(\vec{Y} \times \vec{Y}) + \alpha_y\beta_z(\vec{Y} \times \vec{Z})$$

$$+\alpha_z\beta_x(\vec{Z} \times \vec{X}) + \alpha_z\beta_y(\vec{Z} \times \vec{Y}) + \alpha_z\beta_z(\vec{Z} \times \vec{Z})$$

$$A \times B = \alpha_x\beta_y\vec{Z} - \alpha_x\beta_z\vec{Y} - \alpha_y\beta_x\vec{Z} + \alpha_y\beta_z\vec{X} + \alpha_z\beta_x\vec{Y} - \alpha_z\beta_y\vec{X}$$

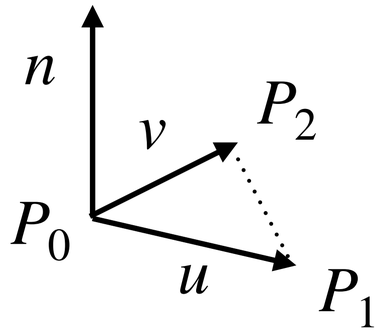
$$A \times B = (\alpha_y\beta_z - \alpha_z\beta_y)\vec{X} + (\alpha_z\beta_x - \alpha_x\beta_z)\vec{Y} + (\alpha_x\beta_y - \alpha_y\beta_x)\vec{Z}$$

Cross Product

Determinant and Matrix form

Determinant form:

$$a \times b = \begin{vmatrix} \vec{X} & \vec{Y} & \vec{Z} \\ \alpha_x & \alpha_y & \alpha_z \\ \beta_x & \beta_y & \beta_z \end{vmatrix} = \begin{bmatrix} \alpha_y\beta_z - \alpha_z\beta_y \\ \alpha_z\beta_x - \alpha_x\beta_z \\ \alpha_x\beta_y - \alpha_y\beta_x \end{bmatrix}$$



Dual Matrix:

$$a \times b = A^{*}b = \begin{bmatrix} 0 & -\alpha_z & \alpha_y \\ \alpha_z & 0 & -\alpha_x \\ -\alpha_y & \alpha_x & 0 \end{bmatrix} \begin{bmatrix} \beta_x \\ \beta_y \\ \beta_z \end{bmatrix}$$

Plane:

$n \cdot (P - P_0) = 0$

where $n = u \times v$

and P, P_0 are points on the plane

Coordinate Frames

Example of dot/cross product use in Computer Graphics

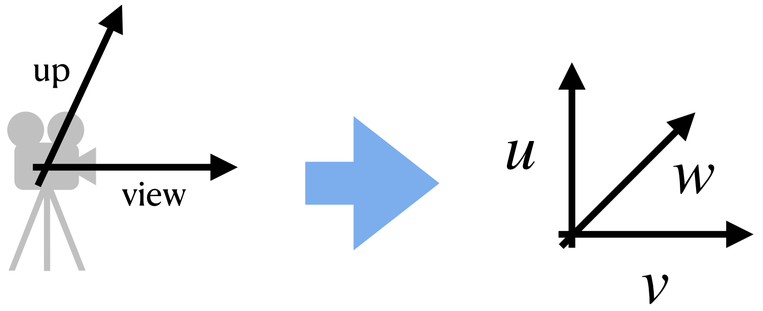
(u, v, w) is a coordinate frame if:

- $|u| = |v| = |w| = 1$
- $u \cdot v = v \cdot w = u \cdot w = 0$
- $w = u \times v$

a point P can be represented by:

$$P = (P \cdot u)u + (P \cdot v)v + (P \cdot w)w$$

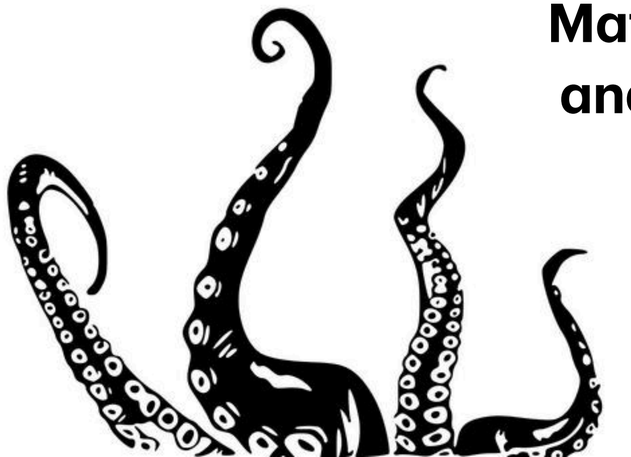
Create a coordinate frame for the **camera** from *view* and *up* vectors.

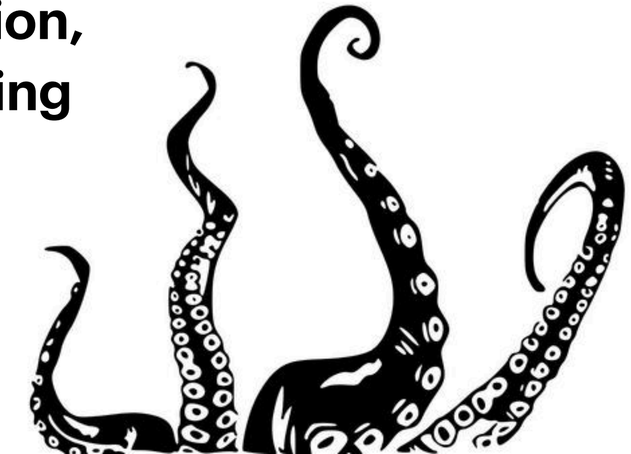


$$v = \frac{view}{|view|}, w = \frac{up \times v}{|up \times v|}, u = v \times w$$

Mathematics for Computer Graphics II

Matrices, Basic Transformations: Scale, Shear, Rotation, and Translation, Homogenous Coordinates, Combining Transformations, Transformation of Normals





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Matrices

Introduction

All transformations in the rendering pipeline are handled by matrices.

Matrices transform points (multiplying a matrix by a vector).

Matrices are arrays of numbers (vectors are 1-column matrices).

Matrix $A_{m,n}$ (m rows and n columns) of elements $a_{i,j}$ (element at row i, column j):

$$A_{m,n} = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{bmatrix}$$

Matrices

Addition and Multiplication

Addition of matrices:

$$A_{m,n} + B_{m,n} = [a_{i,j}] + [b_{i,j}] = [a_{i,j} + b_{i,j}]$$

Multiplication by a scalar:

$$\alpha A_{m,n} = \alpha[a_{i,j}] = [\alpha a_{i,j}]$$

Multiplication by another matrix:

$$\underset{m \times n}{\begin{bmatrix} 1 & 3 \\ 5 & 2 \\ 0 & 4 \end{bmatrix}} \underset{n \times k}{\begin{bmatrix} 3 & 6 & 9 & 4 \\ 2 & 7 & 8 & 3 \end{bmatrix}} = \underset{m \times k}{\begin{bmatrix} 9 & 27 & 33 & 13 \\ 19 & 44 & 61 & 26 \\ 8 & 28 & 32 & 12 \end{bmatrix}}$$

$$\begin{bmatrix} 3 & 6 & 9 & 4 \\ 2 & 7 & 8 & 3 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 5 & 2 \\ 0 & 4 \end{bmatrix} = ?$$

Matrices

Properties

Not commutative: $AB \neq BA$

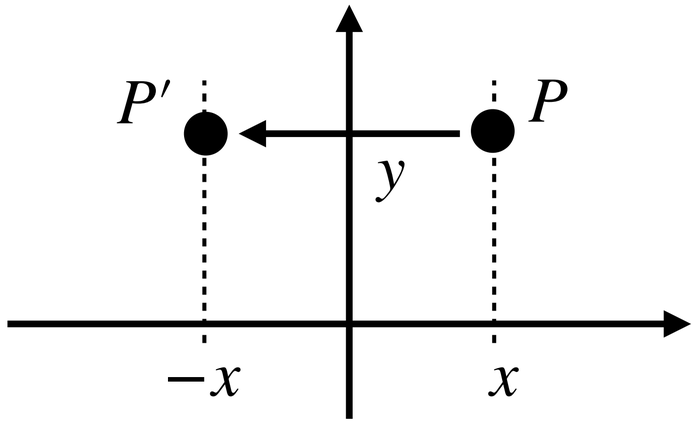
Associative: $(A + B) + C = A + (B + C)$ and $(AB)C = A(BC)$

Distributive: $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$

Matrix-Vector multiplication: $P' = MP$

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -x \\ y \end{bmatrix}$$

M P P'



Matrices

Transpose and Identity

Transpose:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}$$

$$(AB)^T = B^T A^T$$

$$m \times n \quad n \times k \quad k \times n \quad n \times m$$

Additive identity:

$$0 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Multiplicative identity (identity matrix):

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$AA^{-1} = A^{-1}A = I, (AB)^{-1} = B^{-1}A^{-1}$$

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I$$

Matrices

Dot product and Cross product

$$a = \begin{bmatrix} \alpha_x \\ \alpha_y \\ \alpha_z \end{bmatrix} \quad b = \begin{bmatrix} \beta_x \\ \beta_y \\ \beta_z \end{bmatrix}$$

Cross Product:

$$A^* = \begin{bmatrix} 0 & -\alpha_z & \alpha_y \\ \alpha_z & 0 & -\alpha_x \\ -\alpha_y & \alpha_x & 0 \end{bmatrix}$$

Dot Product:

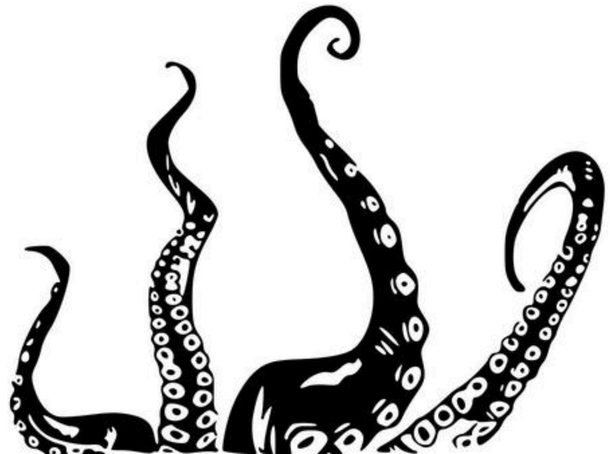
$$A^T B = \begin{bmatrix} \alpha_x & \alpha_y & \alpha_z \end{bmatrix} \begin{bmatrix} \beta_x \\ \beta_y \\ \beta_z \end{bmatrix}$$

$$a \times b = A^* b = \begin{bmatrix} 0 & -\alpha_z & \alpha_y \\ \alpha_z & 0 & -\alpha_x \\ -\alpha_y & \alpha_x & 0 \end{bmatrix} \begin{bmatrix} \beta_x \\ \beta_y \\ \beta_z \end{bmatrix}$$

Basic Transformations

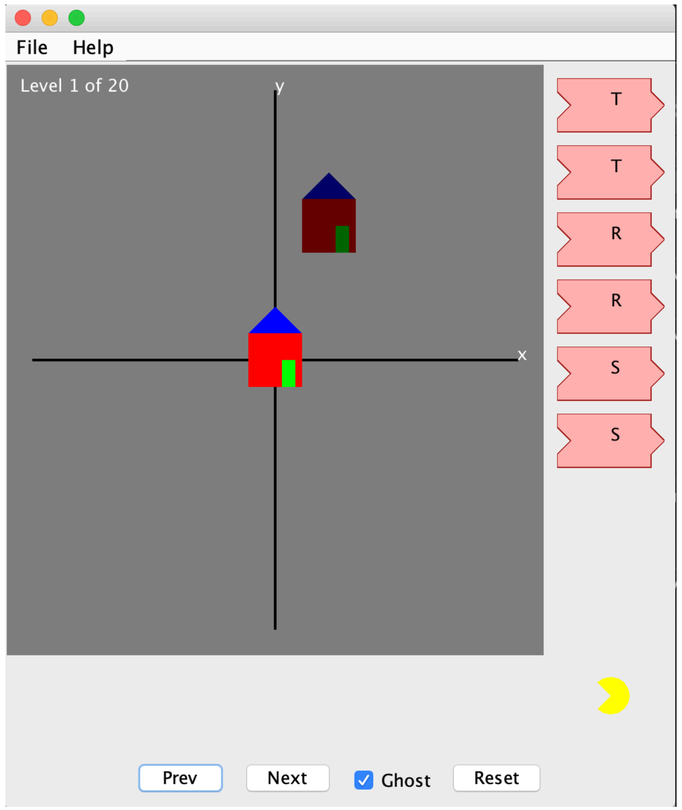
Scale, Shear, 2D and 3D Rotation,
Combining Transformations





Transformation Game

(http://www.cs.brown.edu/exploratories/)

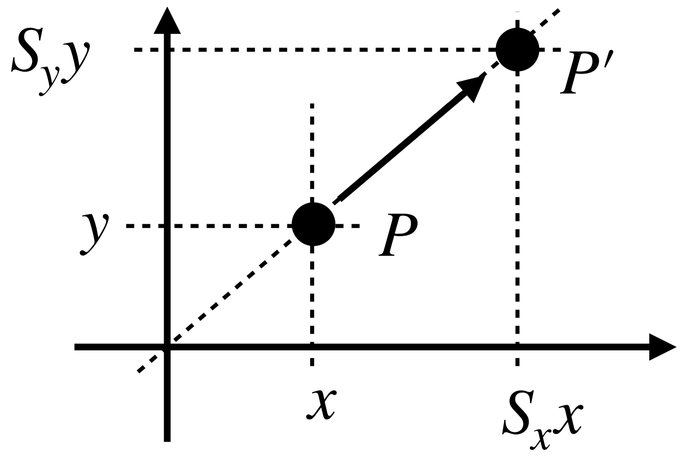


Scale

$$S_{2D}(S_x, S_y) = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix}$$

$$S_{2D}v = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} S_x x \\ S_y y \end{bmatrix}$$

$$S_{2D}^{-1}(S_x, S_y) = \begin{bmatrix} S_x^{-1} & 0 \\ 0 & S_y^{-1} \end{bmatrix}$$

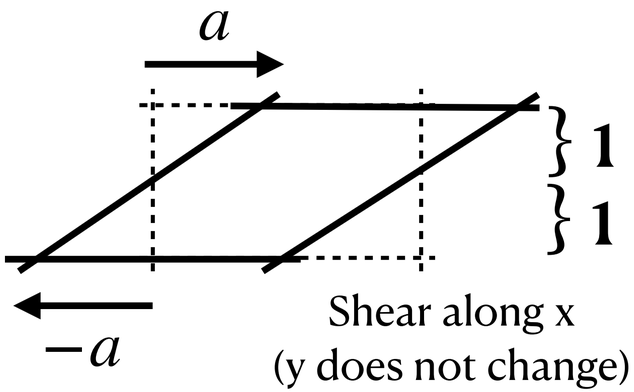


$$S_{3D} = \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & S_z \end{bmatrix}$$

Shear

$$Sh_x(a) = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}, Sh_x v = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + ay \\ y \end{bmatrix}$$

$$Sh_x^{-1}(a) = \begin{bmatrix} 1 & -a \\ 0 & 1 \end{bmatrix}$$



Shear along x
(y does not change)

$y' = y$ and $x' = x + ay$

$$Sh_y(a) = \begin{bmatrix} 1 & 0 \\ a & 1 \end{bmatrix}$$

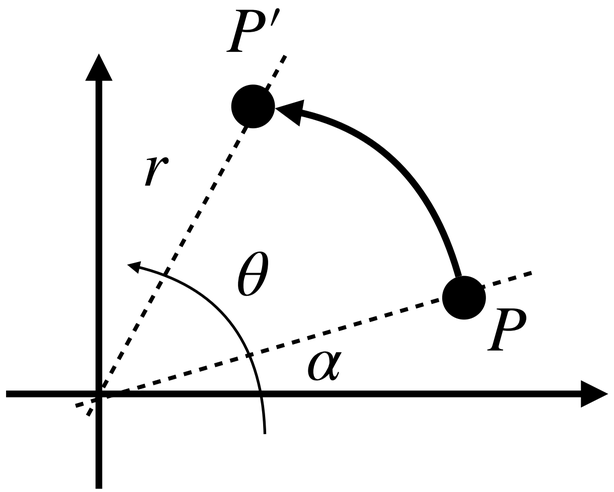
$$Sh_y v = \begin{bmatrix} 1 & 0 \\ a & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ ax + y \end{bmatrix}$$

$$Sh_z v = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ a_{zx} & a_{zy} & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ sh_{zx}x + sh_{zy}y + z \end{bmatrix}$$

etc.

2D Rotation

(commutative)



$$R(x+y) = R(x) + R(y)$$

$$P = (P_x, P_y) = (r\cos\alpha, r\sin\alpha)$$

$$P' = (P'_x, P'_y) = (r\cos(\alpha+\theta), r\sin(\alpha+\theta))$$

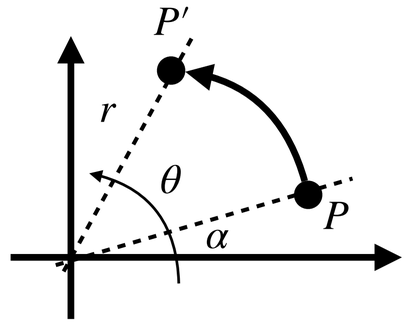
Trigonometric identities:

$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$$

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$$

2D Rotation

$$
\begin{aligned}
P' &= (r\cos(\alpha+\theta), r\sin(\alpha+\theta)) \\
&= (r\cos\alpha\cos\theta - r\sin\alpha\sin\theta, r\sin\alpha\cos\theta + r\cos\alpha\sin\theta) \\
&= (\cos\theta P_x - \sin\theta P_y, cos\theta P_y + \sin\theta P_x)
\end{aligned}
$$



$$
R(\theta)P = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} P_x \\ P_y \end{bmatrix} = \begin{bmatrix} P'_x \\ P'_y \end{bmatrix}
$$

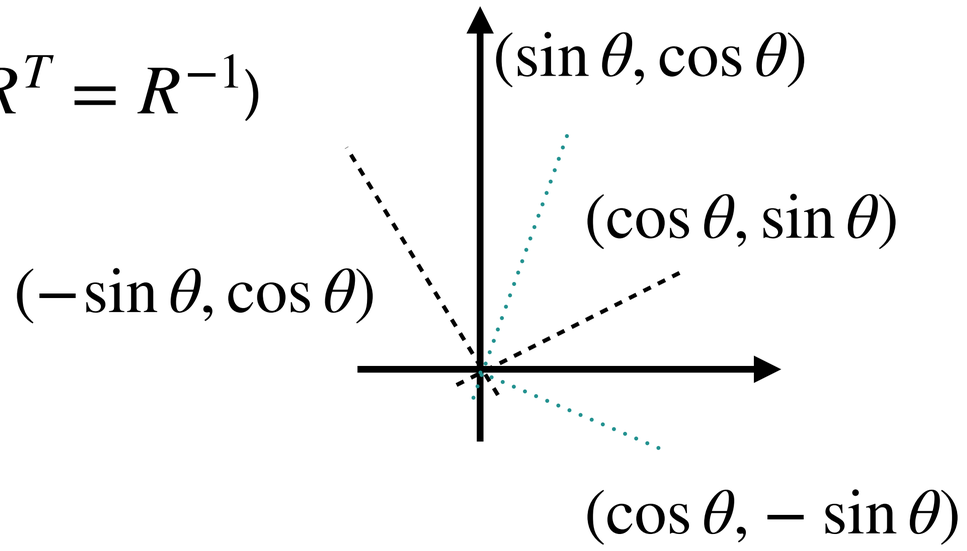
$$
R^{-1}(\theta) = R(-\theta) = \begin{bmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{bmatrix} = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} = R^T(\theta)
$$

2D Rotation

Trigonometric identity: $\cos^2\theta + \sin^2\theta = 1$

$$R(\theta)R^T(\theta) = \begin{bmatrix}\cos(\theta) & -\sin(\theta)\\ \sin(\theta) & \cos(\theta)\end{bmatrix}\begin{bmatrix}\cos(\theta) & \sin(\theta)\\ -\sin(\theta) & \cos(\theta)\end{bmatrix} = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

R is an orthogonal matrix: $R^TR = RR^T = I\ (R^T = R^{-1})$



Combining Transformations

Scale uniformly by a factor of 2 then rotate 45°?

Multiplying matrices representing transformations is still a transformation matrix that represents the composition of the individual transformations.

$x_2 = S(x_1), x_3 = R(x_2) = R(S(x_1)) = (RS)(x_1), x_3 \neq (SR)(x_1)$

How to invert a combination of transforms?

$M = M_1 M_2 M_3 \, , M^{-1} = M_3^{-1} M_2^{-1} M_1^{-1}$

$M^{-1} M = M_3^{-1} M_2^{-1} M_1^{-1} M_1 M_2 M_3 = I$

3D Rotation

Using the cartesian axes

$$R_z = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}, R_z^{-1}\ ?$$

$$R_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}, R_x^{-1}\ ?$$

$$R_y = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}, R_y^{-1}\ ?$$

Matrices are orthogonal and linear.

Remember:

(u, v, w) is a coordinate frame if

$$|u| = |v| = |w| = 1$$

$$u \cdot v = v \cdot w = u \cdot w = 0$$

$$w = u \times v$$

Think of the rows as unit vectors in a new coordinate frame...

3D Rotation

Geometric Interpretation

Think of the rows of the rotation matrix as unit vectors of a new coordinate frame.

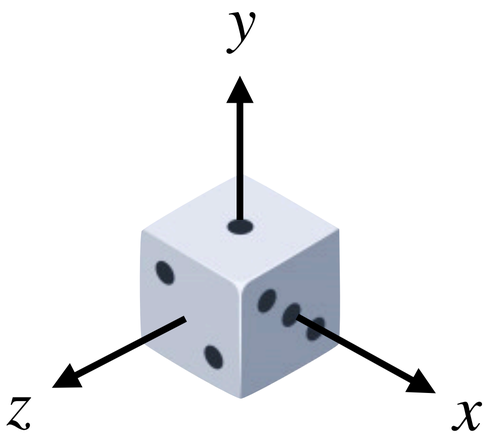
$$R_{uvw}P = \begin{bmatrix} u_x & u_y & u_z \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{bmatrix} \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \begin{bmatrix} u \,.\, P \\ v \,.\, P \\ w \,.\, P \end{bmatrix}$$

We are projecting the vector in the new coordinate frame, e.g. u will become (1,0,0):

$$R_{uvw}u = \begin{bmatrix} u \,.\, u = 1 \\ v \,.\, u = 0 \\ w \,.\, u = 0 \end{bmatrix}, \text{ similarly for the other dimensions.}$$

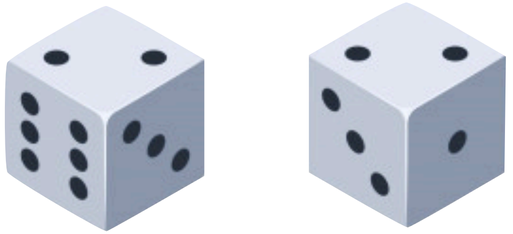
3D Rotation

NOT commutative (unlike 2D rotation)

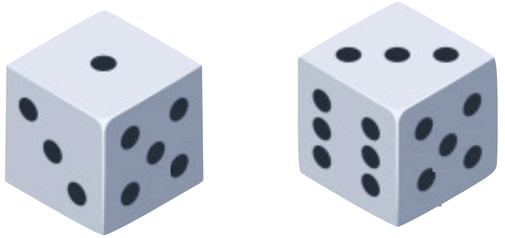


D6

Rotate about x then y:

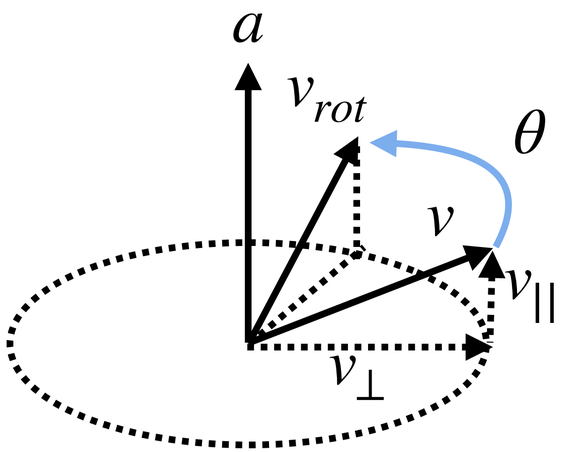


Rotate about y then x:

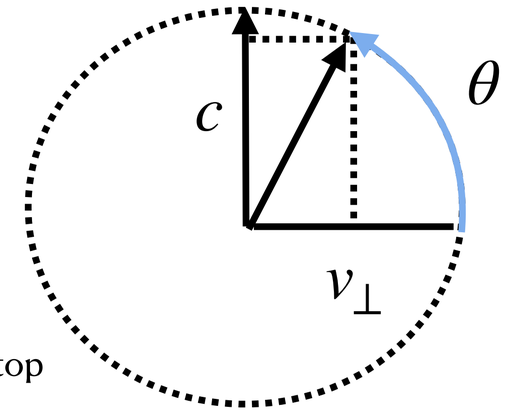


Arbitrary 3D Rotation

about an axis a by an angle of θ



side



top

Assuming $|a| = 1$

Components of v :

- parallel to a : $v_{||} = (a \cdot v)a$ (**projection**)
- orthogonal to a : $v_{\perp} = v - v_{||} = v - (a \cdot v)a$ (**rejection**)

Define $c = a \times v$

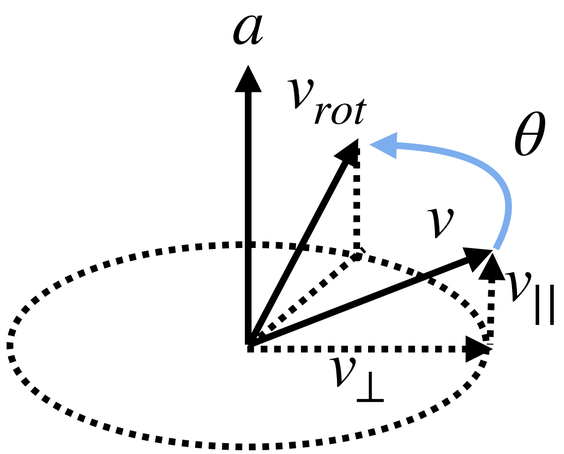
After the rotation, we have:

$$v_{\perp rot} = v_{\perp}\cos\theta + c\sin\theta$$
$$= (v - (a \cdot v)a)\cos\theta + (a \times v)\sin\theta$$

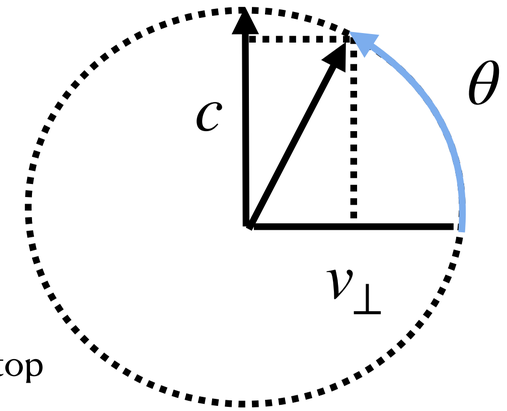
$$v_{||rot} = v_{||} = (a \cdot v)a \text{ (**unchanged**)}$$

Arbitrary 3D Rotation

Rodrigues' Rotation Formula



side



top

$$
\begin{aligned}
v_{rot} &= v_{\perp rot} + v_{||rot} \\
&= (v - (a \cdot v)a)\cos\theta + (a \times v)\sin\theta + (a \cdot v)a \\
&= v\cos\theta - (a\,.\,v)a\cos\theta + (a \times v)\sin\theta + (a \cdot v)a \\
&= v\cos\theta + (a \times v)\sin\theta + a(a \cdot v)(1 - \cos\theta)
\end{aligned}
$$

Rodrigues' Rotation Formula (~1850)

Arbitrary 3D Rotation

Matrix notation of Rodrigues Rotation Formula

$$a = \begin{bmatrix} \alpha_x \\ \alpha_y \\ \alpha_z \end{bmatrix}, v = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}, \text{dual matrix } A = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix} \quad (Av = a \times v)$$

$$\begin{aligned} v_{rot} &= v\cos\theta + (a \times v)\sin\theta + a(a \cdot v)(1 - \cos\theta) \\ &= v - v + v\cos\theta + (a \times v)\sin\theta + a(a \cdot v)(1 - \cos\theta) \\ &= v - v(1 - \cos\theta) + (a \times v)\sin\theta + a(a \cdot v)(1 - \cos\theta) \\ &= v + (Av)\sin\theta + (a(a \cdot v) - v)(1 - \cos\theta) \end{aligned}$$

Arbitrary 3D Rotation

Matrix notation of Rodrigues Rotation Formula

Using vector triple product:

$$i \times (j \times k) = j(i \cdot k) - k(i \cdot j)$$

with $i = j = a$ and $k = v$:

$$a(a \cdot v) - v(a \cdot a) = a \times (a \times v)$$

$$a(a \cdot v) - v = A^2 v$$

$$\begin{aligned} v_{rot} &= v + (Av)\sin\theta + (a(a \cdot v) - v)(1 - \cos\theta) \\ &= v + (\sin\theta)Av + (1 - \cos\theta)A^2 v \\ &= Rv \end{aligned}$$

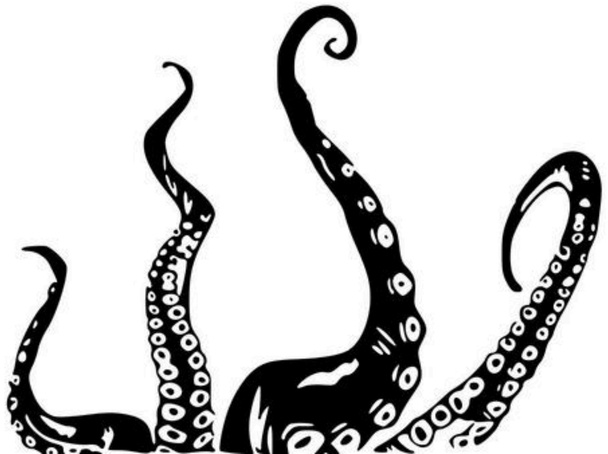
$$R = I + (\sin\theta)A + (1 - \cos\theta)A^2$$

$$A^2 = \begin{bmatrix} -a_y^2 - a_z^2 & a_x a_y & a_x a_z \\ a_x a_y & -a_x^2 - a_z^2 & a_y a_z \\ a_x a_z & a_y a_z & -a_x^2 - a_y^2 \end{bmatrix}$$

Homogeneous Coordinates

Translation, Combining Transformations





Translation

Why leave the easiest for the end?

How to move x by $+7$ units?

$$\begin{bmatrix} ? & ? & ? \\ ? & ? & ? \\ ? & ? & ? \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x+7 \\ y \\ z \end{bmatrix}, T = \begin{bmatrix} ? & ? & ? \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, T = \begin{bmatrix} 1 & 0 & \frac{7}{z} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} ?$$

Wrong: matrices must remain constant and not depend on a point.

Homogeneous Coordinates

Add a 4th coordinate w

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x+7 \\ y \\ z \\ 1 \end{bmatrix}$$

$P = [x \quad y \quad z \quad w]^T$

$w = 0$ represents a point at infinity (vectors), $w > 0$ is a physical point

inhomogeneous point $P = [x \quad y \quad z \quad w]^T = \left[\frac{x}{w} \quad \frac{y}{w} \quad \frac{z}{w} \quad 1\right]^T$

Homogeneous Coordinates

Vectors and 4x4 Transforms

$$\begin{bmatrix} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 0 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \\ 0 \end{bmatrix}$$, as would be expected.

Scaling becomes

$$\begin{bmatrix} S_{3x3} & 0 \\ 0 & 1 \end{bmatrix}$$

Rotation becomes

$$\begin{bmatrix} R_{3x3} & 0 \\ 0 & 1 \end{bmatrix}$$

Translation

$$\begin{bmatrix} I_{3x3} & T_{3x1} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & T_x \\ 0 & 1 & 0 & T_y \\ 0 & 0 & 1 & T_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Combining Transformations

Typically, we take an object, scale it, rotate it, then translate it.

First rotate, then translate:

$$M = TR = \begin{bmatrix} I_{3x3} & T_{3x1} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R_{3x3} & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} R_{3x3} & T_{3x1} \\ 0 & 1 \end{bmatrix}$$

First translate, then rotate:

$$M = RT = \begin{bmatrix} R_{3x3} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} I_{3x3} & T_{3x1} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} R_{3x3} & R_{3x3}T_{3x1} \\ 0 & 1 \end{bmatrix}$$

Homogeneous Coordinates

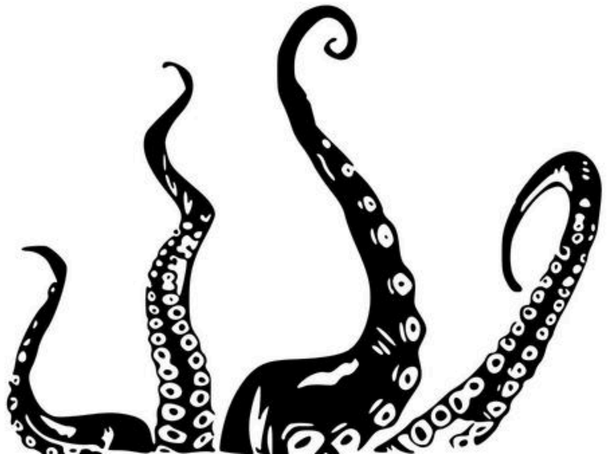
Advantages

- Homogeneous coordinates are a unified framework for transformations
- Can concatenate all transformations in a 4x4 matrix
- Using homogeneous coordinates in perspective viewing requires 1 division
- Simpler formulae that does not blow up (e.g. intersecting parallel lines)
- Standard in Computer Graphics SW & HW

Matrix Inversion

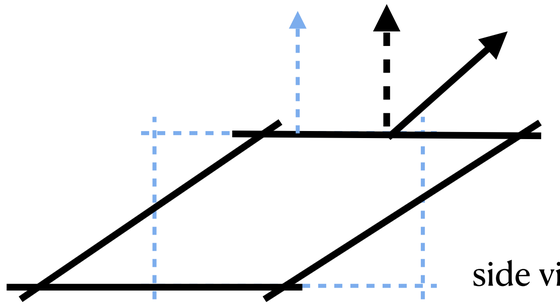
Transforming Normals





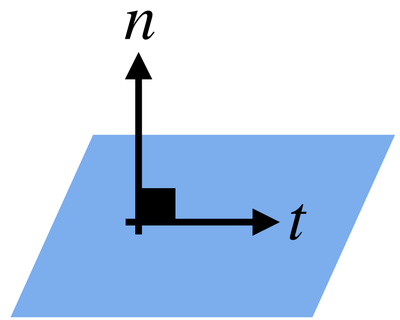
Transforming Normals

Problem:
Tangents are points on the surface and they transform with the object.



incorrect: shear applied to normal

Normal matrix Q:



$$t \to Mt, n \to Qn$$

$$n^T t = 0$$

$$(Qn)^T(Mt) = 0$$

$$(n^T Q^T)(Mt) = 0$$

$$n^T(Q^T M)t = 0 \implies Q^T M = I$$

$$Q = (M^{-1})^T$$

Matrix Inversion

2x2

Determinant $|A|$:

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $|A| = ad - bc$, if $|A| = 0$ then A is not invertible.

Inverse A^{-1}:

$$A^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{|A|} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Matrix Inversion

3x3 through an example

$$M = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}$$

Determinant $|M|$:

$$\begin{vmatrix} + & - & + \\ 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{vmatrix} \quad \begin{vmatrix} + & - & + \\ 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{vmatrix} \quad \begin{vmatrix} + & - & + \\ 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{vmatrix}$$

$$\begin{aligned} |M| &= +1 \times (1 \times 0 - 4 \times 6) \\ &\quad -2 \times (0 \times 0 - 5 \times 4) \\ &\quad +3 \times (0 \times 6 - 1 \times 5) \\ &= -24 + 40 - 15 = 1 \end{aligned}$$

Matrix Inversion

determinant of 3x3 matrix

$$M = \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{bmatrix}$$

$$\begin{aligned} |M| = & \, m_{11}(m_{22}m_{33} - m_{23}m_{32}) \\ & - m_{12}(m_{21}m_{33} - m_{23}m_{31}) \\ & + m_{13}(m_{21}m_{32} - m_{22}m_{31}) \end{aligned}$$

$$M = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}$$

Matrix Inversion

inverse of 3x3 matrix

1. **Determinant** $|M|$

$$|M| = 1 \neq 0$$

2. **Transpose** M^T

$$M^T = \begin{bmatrix} 1 & 0 & 5 \\ 2 & 1 & 6 \\ 3 & 4 & 0 \end{bmatrix}$$

3. **Determinant of minor matrices**

$$\begin{vmatrix} 1 & 6 \\ 4 & 0 \end{vmatrix} = -24, \begin{vmatrix} 2 & 6 \\ 3 & 0 \end{vmatrix} = -18, \begin{vmatrix} 2 & 1 \\ 3 & 4 \end{vmatrix} = 5,$$

$$\begin{vmatrix} 0 & 5 \\ 4 & 0 \end{vmatrix} = -20, \begin{vmatrix} 1 & 5 \\ 3 & 0 \end{vmatrix} = -15, \begin{vmatrix} 1 & 0 \\ 3 & 4 \end{vmatrix} = 4,$$

$$\begin{vmatrix} 0 & 5 \\ 1 & 6 \end{vmatrix} = -5, \begin{vmatrix} 1 & 5 \\ 2 & 6 \end{vmatrix} = -4, \begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} = 1$$

$$M = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}$$

Matrix Inversion

inverse of 3x3 matrix

4. Matrix of cofactors / Adjugate / Adjoint

$$\begin{bmatrix} -24 & -18 & 5 \\ -20 & -15 & 4 \\ -5 & -4 & 1 \end{bmatrix} \sim> \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix} = \begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{bmatrix} = Adj(M)$$

5. $$M^{-1} = \frac{1}{|M|} Adj(M) = \begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{bmatrix}$$

Thank You

for your attention!

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